Kernix 0.5.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

18 workloads

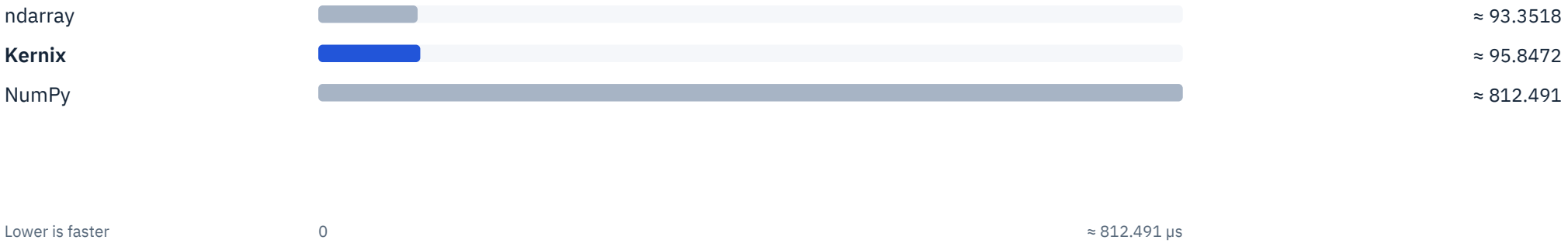
53 observations

Source precision retained

Matrix multiplication

Batch-mean median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Historical medians of 15 batch means. Recorded p95 is the batch maximum.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

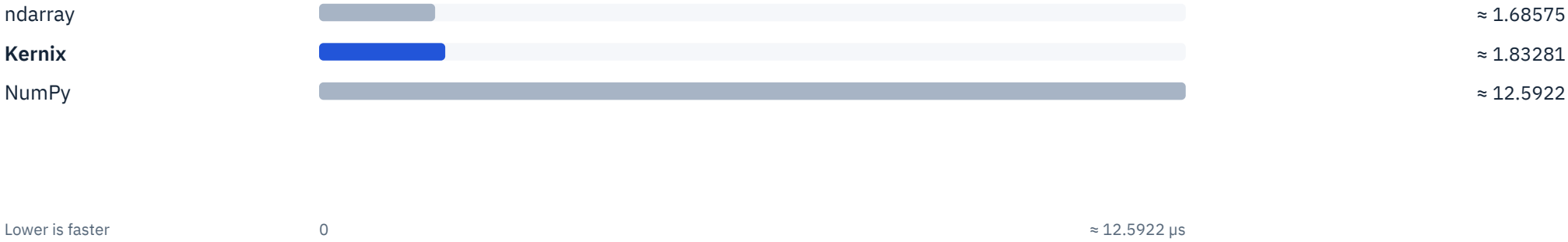
Kernix 0.5.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Matrix multiplication

Batch-mean median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



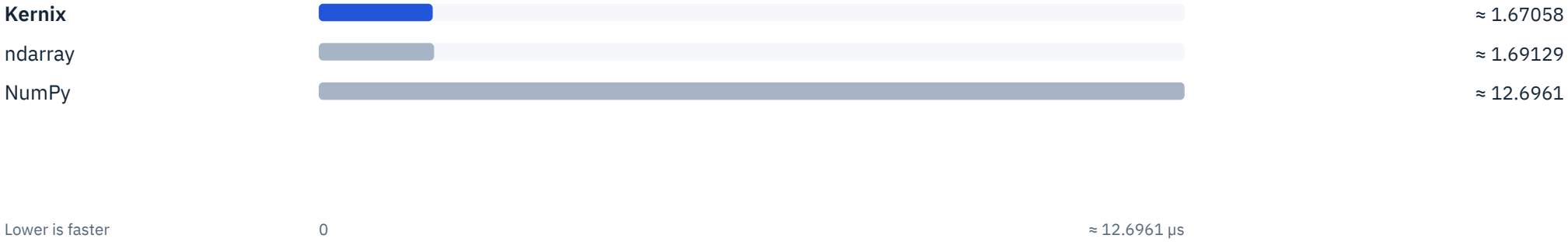
Kernix 0.5.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Matrix multiplication · reused output

Batch-mean median

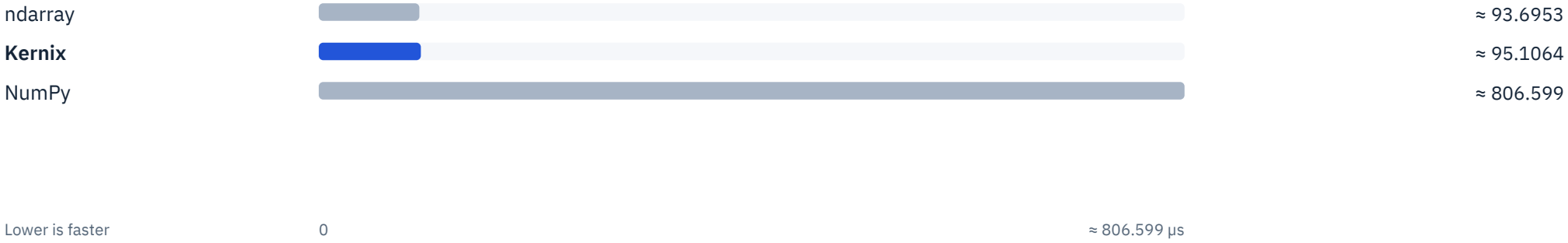
32 × 32 · 32 × 32 · f64 · 1 worker · reused output



Matrix multiplication · reused output

Batch-mean median

128 × 128 · 128 × 128 · f64 · 1 worker · reused output



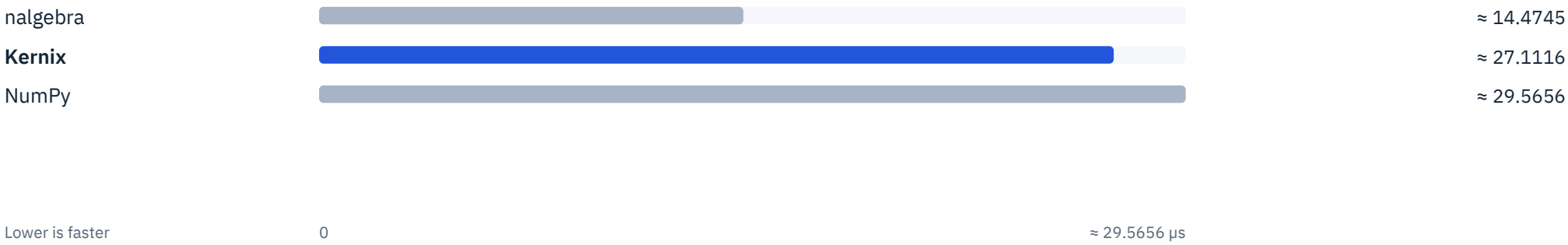
Kernix 0.5.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Cholesky factorization

Batch-mean median

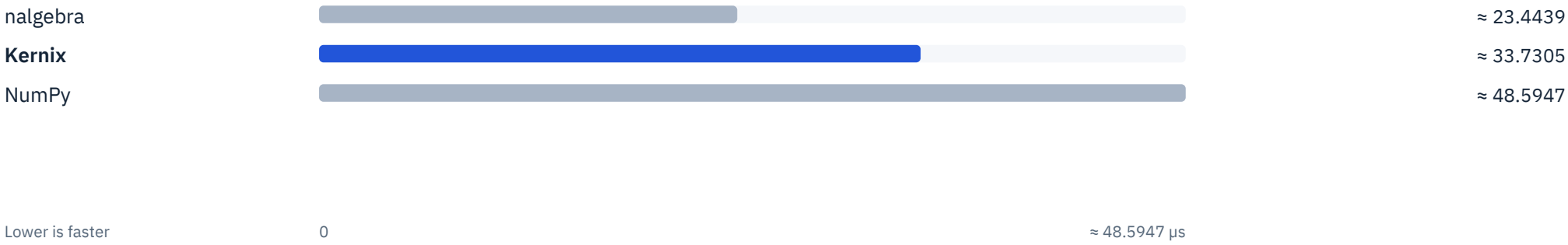
64 × 64 · f64 · 1 worker · allocated output



Linear system solve

Batch-mean median

64 × 64 · f64 · 1 worker · allocated output



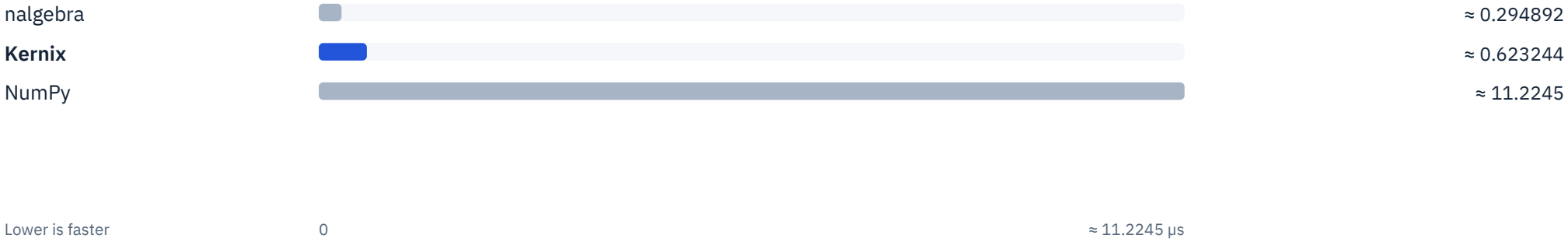
Kernix 0.5.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Thin QR

Batch-mean median

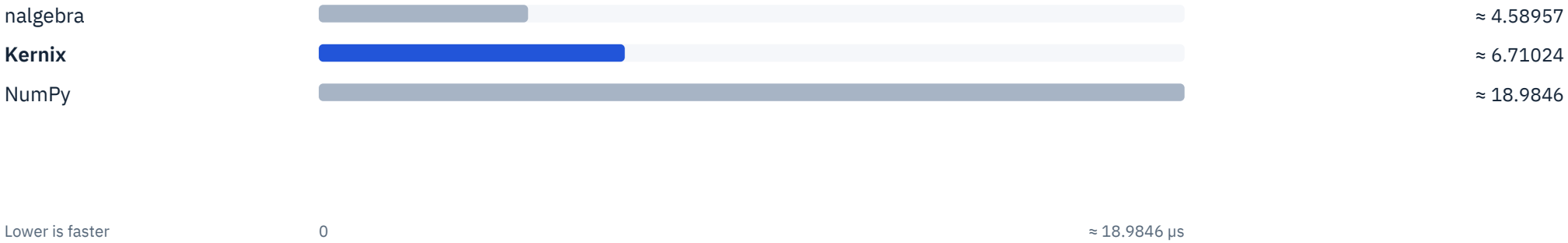
8 × 4 · f64 · 1 worker · allocated output



Thin QR

Batch-mean median

32 × 16 · f64 · 1 worker · allocated output



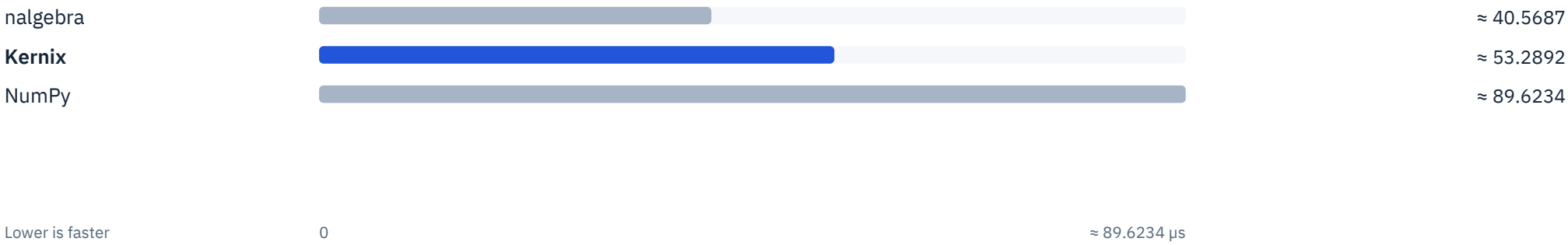
Kernix 0.5.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Thin QR

Batch-mean median

96 × 32 · f64 · 1 worker · allocated output



Thin QR

Batch-mean median

256 × 64 · f64 · 1 worker · allocated output



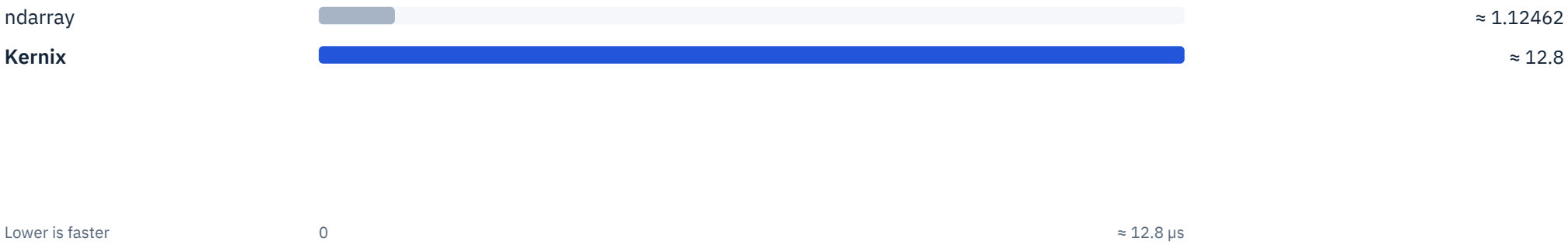
Kernix 0.5.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Dense training step

Batch-mean median

32 × 16 × 8 · f64 · Graph-based Kernix vs hand-derived comparator updates; API costs differ



Kernix 0.5.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Elementwise addition

Batch-mean median

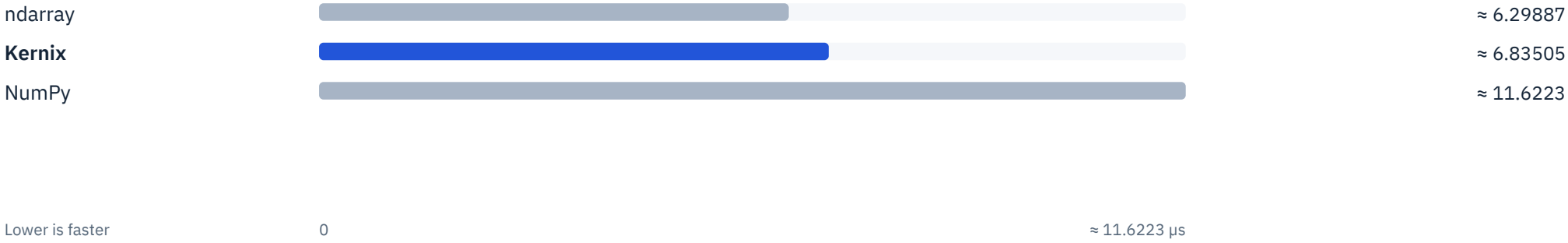
65,536 elements · f64 · 1 worker · allocated output



Sum reduction

Batch-mean median

65,536 elements · f64 · 1 worker · scalar output



Kernix 0.5.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

Batch-mean median



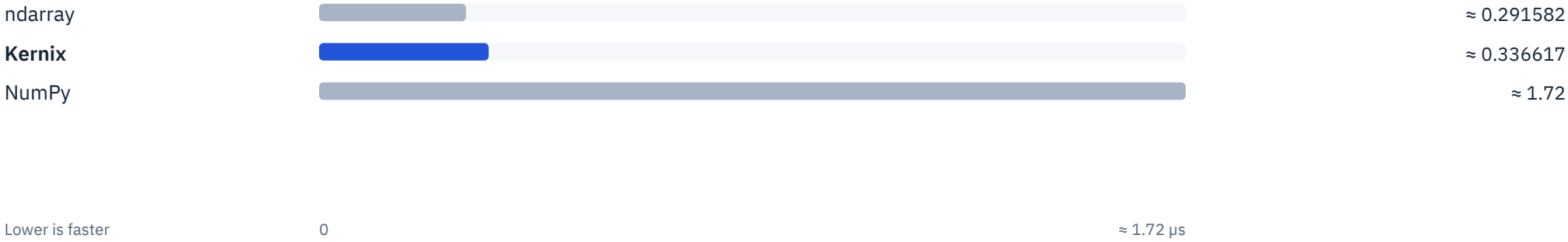
Kernix 0.5.0 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Create filled tensor

Batch-mean median

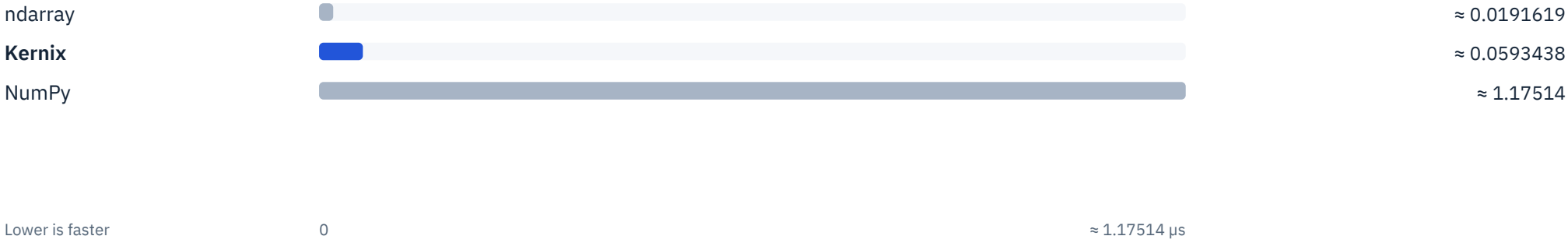
64 × 64 · f64 · 1 worker · allocated output



Create matrix

Batch-mean median

3 × 3 · f64 · 1 worker · allocated output



Kernix 0.5.0 / Vision workloads

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

RGB resize & normalization

Batch-mean median

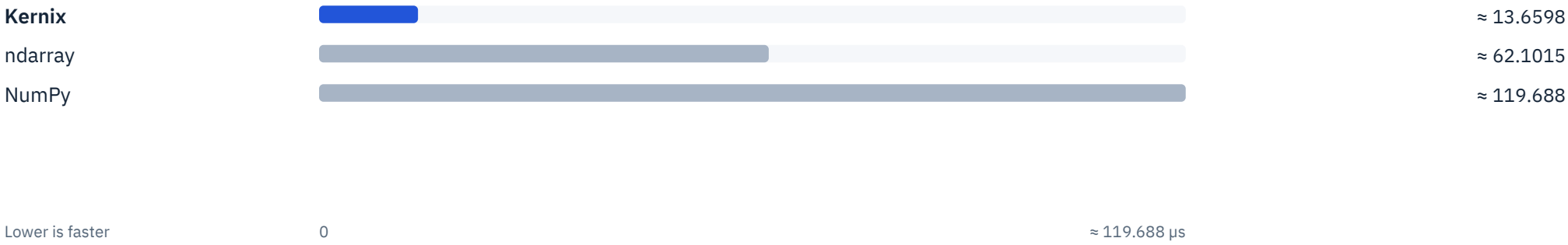
1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32 · 1 worker · allocated output · comparator implementations differ



2D convolution

Batch-mean median

1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32 · 1 worker · allocated output · comparator implementations differ



Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1832.807931969797	1.832807931969797
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1685.7492414537119	1.6857492414537119
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	12592.181772406848	12.592181772406848
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	95847.16858237548	95.84716858237548
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	93351.80970149254	93.35180970149254
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	812491.2580645161	812.4912580645161
Matrix multiplication · reused output / Kernix 32 × 32 · 32 × 32 · f64	1670.5828265953892	1.6705828265953892
Matrix multiplication · reused output / ndarray 32 × 32 · 32 × 32 · f64	1691.2876276804438	1.6912876276804438
Matrix multiplication · reused output / NumPy 32 × 32 · 32 × 32 · f64	12696.077664974619	12.696077664974619
Matrix multiplication · reused output / Kernix 128 × 128 · 128 × 128 · f64	95106.36121673005	95.10636121673005

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication · reused output / ndarray 128 × 128 · 128 × 128 · f64	93695.25842696629	93.69525842696629
Matrix multiplication · reused output / NumPy 128 × 128 · 128 × 128 · f64	806599.0322580645	806.5990322580645
Cholesky factorization / Kernix 64 × 64 · f64	27111.61863488624	27.11161863488624
Cholesky factorization / nalgebra 64 × 64 · f64	14474.516203703704	14.474516203703704
Cholesky factorization / NumPy 64 × 64 · f64	29565.643026004727	29.565643026004727
Linear system solve / Kernix 64 × 64 · f64	33730.46226415094	33.73046226415094
Linear system solve / nalgebra 64 × 64 · f64	23443.933458294283	23.443933458294283
Linear system solve / NumPy 64 × 64 · f64	48594.65631067961	48.59465631067961
Thin QR / Kernix 8 × 4 · f64	623.2443098247451	0.6232443098247451
Thin QR / nalgebra 8 × 4 · f64	294.89248263090224	0.29489248263090224

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / NumPy 8 × 4 · f64	11224.477109515261	11.224477109515261
Thin QR / Kernix 32 × 16 · f64	6710.243692968331	6.710243692968331
Thin QR / nalgebra 32 × 16 · f64	4589.567364170338	4.589567364170338
Thin QR / NumPy 32 × 16 · f64	18984.610478359908	18.984610478359908
Thin QR / Kernix 96 × 32 · f64	53289.24042553191	53.28924042553191
Thin QR / nalgebra 96 × 32 · f64	40568.70664505673	40.56870664505673
Thin QR / NumPy 96 × 32 · f64	89623.41935483871	89.62341935483871
Thin QR / Kernix 256 × 64 · f64	573412.4318181818	573.4124318181818
Thin QR / nalgebra 256 × 64 · f64	451608.125	451.608125
Thin QR / NumPy 256 × 64 · f64	810888.2580645161	810.8882580645161

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Dense training step / Kernix 32 × 16 × 8 · f64	12799.961105424769	12.799961105424769
Dense training step / ndarray 32 × 16 × 8 · f64	1124.6242465137202	1.1246242465137202
Elementwise addition / Kernix 65,536 elements · f64	28160.966216216217	28.160966216216217
Elementwise addition / ndarray 65,536 elements · f64	25570.189161554194	25.570189161554194
Elementwise addition / NumPy 65,536 elements · f64	25859.510858324717	25.859510858324717
Sum reduction / Kernix 65,536 elements · f64	6835.045379989065	6.835045379989065
Sum reduction / ndarray 65,536 elements · f64	6298.869992441421	6.298869992441421
Sum reduction / NumPy 65,536 elements · f64	11622.313197026022	11.622313197026022
Row softmax / Kernix 256 × 64 · f64	95105.27376425856	95.10527376425856
Row softmax / ndarray 256 × 64 · f64	84504.88175675676	84.50488175675676

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Row softmax / NumPy 256 × 64 · f64	113751.11363636363	113.75111363636363
Create filled tensor / Kernix 64 × 64 · f64	336.61695996983934	0.33661695996983934
Create filled tensor / ndarray 64 × 64 · f64	291.582236995568	0.291582236995568
Create filled tensor / NumPy 64 × 64 · f64	1720.0022702256467	1.7200022702256467
Create matrix / Kernix 3 × 3 · f64	59.34381104978933	0.05934381104978933
Create matrix / ndarray 3 × 3 · f64	19.161851430008017	0.019161851430008017
Create matrix / NumPy 3 × 3 · f64	1175.144394829612	1.175144394829612
RGB resize & normalization / Kernix 1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32	573215.1590909091	573.2151590909091
RGB resize & normalization / ndarray 1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32	768795.9393939395	768.7959393939395
RGB resize & normalization / NumPy 1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32	4425306.333333333	4425.306333333333

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
2D convolution / Kernix 1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32	13659.844347351174	13.659844347351174
2D convolution / ndarray 1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32	62101.503722084366	62.101503722084366
2D convolution / NumPy 1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32	119687.59330143541	119.68759330143541